Chapter 5

Attenuation analysis

The spatial attenuation of surface waves as they propagate away from the source is associated with the geometric spreading and with the intrinsic energy dissipation caused by material damping. The estimation of the intrinsic attenuation component can be used for the characterization of the dissipative properties of the medium, typically assuming a viscoelastic constitutive model. This chapter deals with the methods used to estimate the surface wave attenuation parameters to be used in a subsequent inversion process aimed at estimating the small-strain damping ratio profile (see Chapter 6).

Attenuation analysis is generally performed on multichannel measurements acquired using a linear array of receivers and an active source. The procedure requires accurate measurements of the amplitude of the surface wave particle motion. It is essential that the effects of the noise are considered and that the amplitude perturbations are minimized. For example, the verticality and physical coupling of each receiver should be checked carefully. Moreover, an accurate calibration of the receivers is needed to guarantee a uniform response by the array. If methods based on the source force are used, the receiver sensitivity is required for the conversion of the electrical signal to physical units, and it has to be specifically evaluated in the calibration process.

5.1 ATTENUATION OF SURFACE WAVES

A complete discussion of surface wave propagation in viscoelastic media is presented in Chapter 2. In this section, we briefly review the main parameters related to the attenuation of surface waves. The frequency-dependent complex wavenumber k_n^* defines the propagation of dispersive surface wave modes in linear viscoelastic media. The phase velocity and the attenuation constant for the nth mode are related to the complex wavenumber by

$$k_n^* = Re\left(k_n^*\right) + Im\left(k_n^*\right) = \left(k_n - i\alpha_n\right) = \left(\frac{\omega}{v_n} - i\alpha_n\right) \tag{5.1}$$

Table 5.1 Properties of layered system used to generate the examples of Figures 5.1 and 5.2

Layer	*Thickness (m)*	*Vp (m/s)*	*Vs (m/s)*	*Dp (–)*	*Ds (–)*	*ρ (Mg/m³)*
1	5	400	200	0.04	0.04	1.9
2	5	500	250	0.03	0.03	1.9
3	5	600	300	0.025	0.025	1.9
Half-space	∞	800	400	0.02	0.02	1.9

The real part of the wavenumber is the physical wavenumber, a function of the real physical phase velocity v_n; the imaginary part of the complex wavenumber is the attenuation constant. The frequency-dependent attenuation constant α_n is the coefficient of the exponential function describing the spatial decay of the spectral amplitude for a single mode in the far-field

$$A_n(\omega, r) = A_{n,0}(\omega) \cdot \frac{1}{\sqrt{r}} \cdot e^{-\alpha_n(\omega) \cdot r} \tag{5.2}$$

where r is the source to receiver distance. To show the relationship between complex wavenumber, velocity, and damping, let us consider the simple layered medium of Table 5.1.

The solution of the Rayleigh eigenvalue problem for the medium of Table 5.1 is obtained following the procedure detailed in Chapter 2. The real and imaginary parts of the complex wavenumber are plotted in Figure 5.1.

Relating the complex wavenumber to the material properties is not a straightforward process. Indeed, even in a medium with homogeneous velocity and intrinsic absorption, both wavenumber and attenuation coefficient are frequency dependent. A representation showing the relationship with the material properties can be obtained using two different parameters, derived from the real and imaginary part of the wavenumber. The first one is the modal phase velocity, which is $v_n = \frac{\omega}{Re(k_n)}$. It has a more intuitive and direct relationship with the layer velocities.

The second parameter is the Rayleigh modal *phase damping ratio* (Misbah and Strobbia 2014), which is defined using an analogy with the body wave damping ratio in a viscoelastic medium, as

$$D(\omega) = \frac{Im\left(k^{*2}\right)}{2 \cdot Re\left(k^{*2}\right)} \tag{5.3}$$

The Rayleigh phase damping ratio is a modal property, related to the damping ratio of the subsurface layers. The same information in Figure 5.1 is represented in terms of phase velocity and phase damping ratio in Figure 5.2.

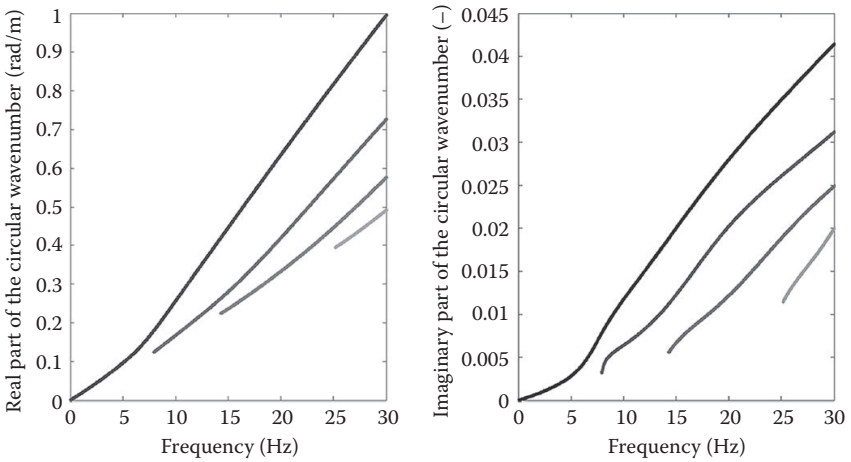

Figure 5.1 Real and imaginary parts of the complex wavenumber of the first four modes for the model of Table 2.1.

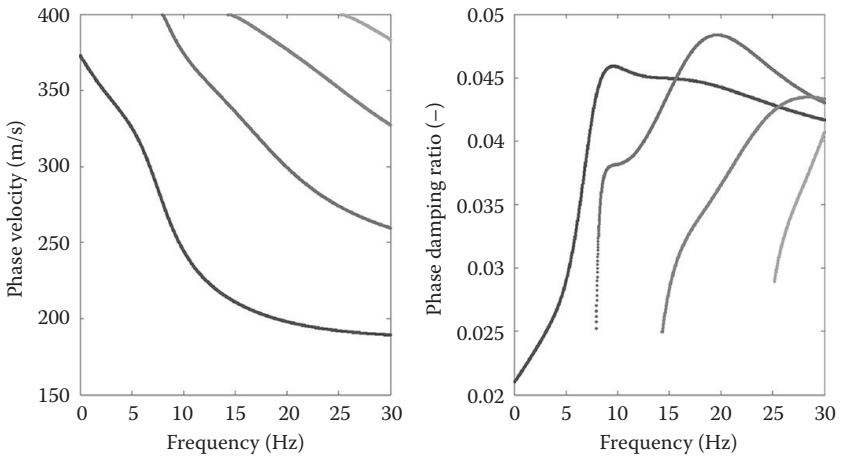

Figure 5.2 Phase velocity and phase damping ratio curves of the first four modes for the model of Table 2.1.

The variation of damping as a function of frequency is related to the geometric dispersion.

The two sets of parameters (the real and imaginary parts of the complex wavenumber, or phase velocity and phase damping) are equivalent. Nevertheless, the phase damping curves allow a more intuitive assessment and evaluation of attenuation data.

As an alternative to the phase damping ratio, it is possible to define a phase quality factor Q, related to the subsurface quality factor, as

$$Q(\omega) = \frac{Re\left(k^{*2}\right)}{Im\left(k^{*2}\right)} \tag{5.4}$$

In the following sections, most results will be presented in terms of attenuation constant curves, although the phase damping ratio will be used in Section 5.4.

5.2 UNIVARIATE REGRESSION OF AMPLITUDE VERSUS OFFSET DATA

Regression of the amplitude of particle motion as a function of offset from the source is the straightforward method for the evaluation of attenuation coefficients. With no loss of generality, in the following, reference is made to vertical particle motion on the ground surface $u_2(r,t)$, which is a function of distance from the source r. Because the analysis is performed in the frequency domain, the experimental data have to be expressed in terms of spectral quantities.

When geophones are used as receivers, the relevant experimental spectral quantity is the particle velocity auto power spectrum $G_{rr}(\omega)$, which is calculated at each receiver location. The experimental particle displacement spectrum $U_2(r,\omega)$ can be then readily computed as

$$\left|U_2(r,\omega)\right| = \frac{\left|\dot{U}_2(r,\omega)\right|}{\omega \cdot C(\omega)} = \frac{\sqrt{G_{rr}(\omega)}}{\omega \cdot C(\omega)} \tag{5.5}$$

where $C(\omega)$ is a frequency-dependent calibration factor that converts the electrical output of the velocity transducer (current, e.g., V) into kinematical units (velocity, e.g., m/s). Equation 5.5 can be easily modified for the use of accelerometers, instead of geophones, as receivers.

Once the particle displacement amplitudes $\left|U_2(r,\omega)\right|$ have been determined, the Rayleigh attenuation coefficients $\alpha_R(\omega)$ can be computed from a nonlinear regression based on the equation that expresses particle motion as a function of source offset for a harmonic excitation (see Section 2.4.2)

$$\left|U_2(r,\omega)\right| = F \cdot Y(r,\omega) \cdot e^{-\alpha_R(\omega)\cdot r} \tag{5.6}$$

where $Y(r,\omega)$ is the Rayleigh geometric spreading function describing the geometric attenuation of multimode Rayleigh waves in vertically heterogeneous media, and F is the magnitude of the vertical harmonic source applied

at the free surface. The geometric spreading function can be calculated from the solution of the Rayleigh forward problem, if the shear wave velocity profile of the site is available (Rix et al. 2000). The latter can be determined from the inversion of the experimental dispersion curve or assumed from other available data. The experimental attenuation coefficients $\alpha_R(\omega)$ determined with the regression process are apparent attenuation coefficients that may represent the combination of several modes of Rayleigh wave propagation.

Examples of regression of particle displacement spectra are provided in Figure 5.3. As the frequency increases, the effects of mode superposition on geometric attenuation produce marked oscillations in the theoretical trend of displacements as a function of offset. This aspect is particularly pronounced in the examples of Figure 5.3 because the site is inversely

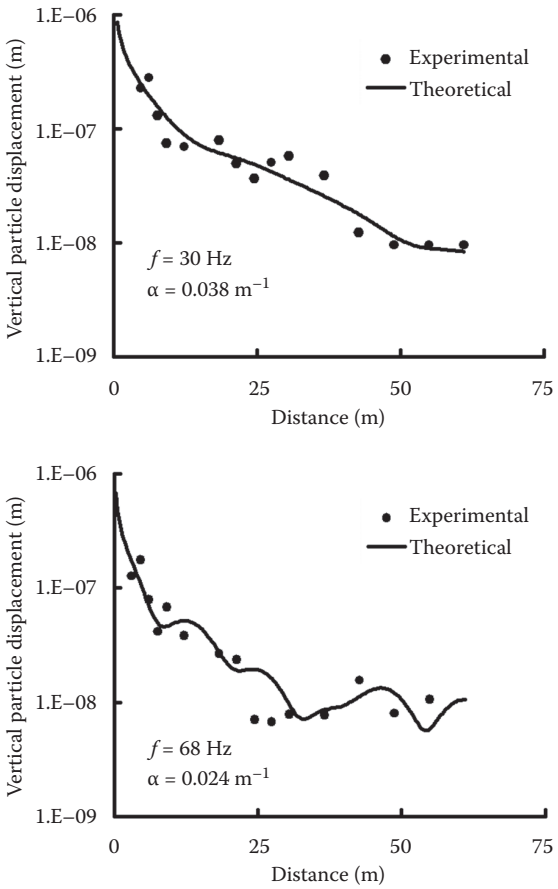

Figure 5.3 Examples of regression of vertical particle displacement amplitude versus offset. (From Rix, G. J. et al., *J Geotech Geoenviron Eng*, 126, 472–480, 2000.)

dispersive with a stiff crust over softer materials (Rix et al. 2000). Moreover, the oscillations in the geometric spreading function multiply for increasing frequency. For this reason, the accuracy in the evaluation of the attenuation coefficients is strongly dependent on the accuracy of the available shear wave velocity profile, especially for the high-frequency values.

A simplified approach can be devised assuming that the propagation of surface waves is dominated by the fundamental mode of Rayleigh wave propagation. Under this assumption, the geometric attenuation is described by the factor associated with surface wave propagation in homogeneous media, which is proportional to the inverse of the square root of the offset from the source. The equation of motion of surface wave propagation (Equation 5.6) simplifies into

$$\left|U_2(r,\omega)\right| = F \cdot \frac{e^{-\alpha_R(\omega)\cdot r}}{\sqrt{r}} \tag{5.7}$$

An example of regression of particle displacement spectra with Equation 5.7 is reported in Figure 5.4. Clearly in this case, the theoretical particle displacements follow a much simplified pattern. Errors associated with this simplification grow whenever higher modes play a relevant role in the propagation and in the increase of frequency.

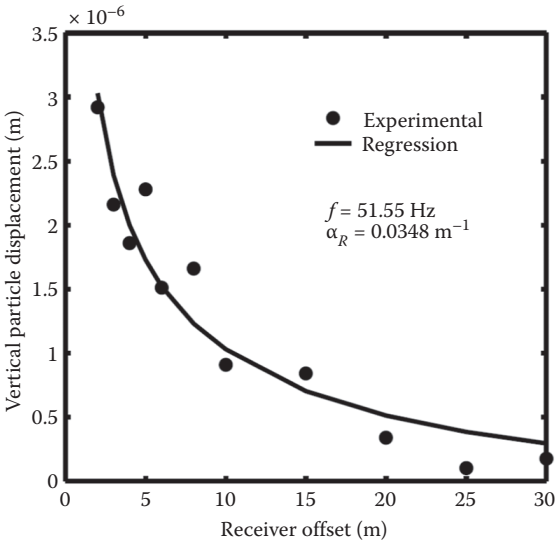

Figure 5.4 Examples of regression of vertical particle displacement amplitude versus offset with simplified expression for the geometric attenuation. (From Rix G. J. et al., *Geotech Test J*, 24, 350–358, 2001.)

In order to avoid uncertainties associated with the coupling of the source to the ground, the regression process also can be performed considering the amplitude of the input force F as an unknown variable. This also allows a straightforward extension of the technique to experimental data collected from sources other than controlled harmonic ones (e.g., impulsive sources), which are cheaper and more time efficient.

This procedure assumes that the particle motion on the ground surface is only caused by surface wave propagation, neglecting body wave contributions; hence, the position of receivers should be selected in order to minimize near-field effects, which are not taken into account in the analysis.

5.3 TRANSFER FUNCTION TECHNIQUE AND COMPLEX WAVENUMBERS

Dispersion and attenuation of surface waves are two aspects of the same phenomenon; hence, a robust and elegant approach can be devised to obtain the two pieces of information simultaneously. In Section 4.4, the regression of phase versus offset was introduced as a method to evaluate dispersion of surface waves, whereas in Section 5.2, the regression of particle amplitudes has been used to analyze surface wave attenuation. The two approaches can be unified by introducing the concept of displacement transfer function, which allows for the experimental dispersion and attenuation curves to be evaluated simultaneously using a single set of measurements. The procedure is then complemented by the coupled inversion that will be introduced in Section 6.4.3 to get a robust and consistent framework for fully coupled analysis of surface wave data leading to the simultaneous estimate of shear wave velocity and damping ratio profiles (see also the examples in Section 7.3).

In a linear system, which in this case is a linear viscoelastic medium, the ratio between the output and the input signals in the frequency domain is called the frequency response function, or transfer function, of the system (Oppenheim and Willsky 1997). In the case of surface wave testing, the input signal is the harmonic force $F \cdot e^{i\omega t}$ applied by the source (e.g., a vertically oscillating shaker) and the output signal is the vertical displacement on the ground surface $U_2(r,\omega)$ measured at a distance r from the source.

For far-field measurements, the vertical displacement $u_2(r,\omega)$ in a linear viscoelastic vertically heterogeneous medium by a harmonic source $F \cdot e^{i\omega t}$ located at the ground surface can be written as (see Section 2.4.2)

$$u_2(r,\omega) = F \cdot Y(r,\omega) \cdot e^{i[\omega t - \Psi(r,\omega)]} \tag{5.8}$$

where $\Psi(r,\omega)$ is the complex-valued phase angle and $Y(r,\omega)$ is the geometrical spreading function. Hence, the displacement transfer function $T(r,\omega)$ between the harmonic source and the receiver is given by

$$T(r,\omega)=\frac{u_2(r,\omega)}{F\cdot e^{i\omega t}}=Y(r,\omega)\cdot e^{-i\cdot\Psi(r,\omega)} \tag{5.9}$$

Assuming $\Psi(r,\omega)\approx k^*(\omega)\cdot r$, where $k^*(\omega)=\left[\frac{\omega}{V_R(\omega)}+i\alpha_R(\omega)\right]$ is the complex wavenumber, the implicit dependence of the complex-valued phase angle on the source-to-receiver distance is eliminated. Equation 5.9 thus becomes

$$T(r,\omega)=\frac{u_2(r,\omega)}{F\cdot e^{i\omega t}}=Y(r,\omega)\cdot e^{-i\cdot k^*(\omega)\cdot r} \tag{5.10}$$

Because $T(r,\omega)$ can be directly measured on site if the input source is adequately monitored, Equation 5.10 can be used as a basis of a nonlinear regression analysis for determining the complex wavenumber $k^*(\omega)$. The optimization procedure can be implemented with a classical least-squares technique; however, because $k^*(\omega)$, $Y(r,\omega)$, and $T(r,\omega)$ are complex-valued quantities, the actual implementation of the algorithm requires the definition of norm valid in a pre-Hilbert space (Parker 1994).

The assumption $\Psi(r,\omega)\approx k^*(\omega)\cdot r$ is equivalent to considering the phase angle $\Psi(r,\omega)$ as the result of a single mode of propagation. As a consequence, the method determines apparent values of Rayleigh phase velocities and attenuation coefficients, which can be influenced by modal superposition if there is not a single dominant mode (see Section 2.4.2).

The procedure requires knowledge of the geometric spreading function $Y(r,\omega)$, which can be calculated if a shear wave velocity model of the site is available. An iterative procedure is therefore necessary. At first, the experimental dispersion and attenuation curves are obtained assuming the geometric spreading function $Y(r,\omega)$ to be proportional to $1/\sqrt{r}$, such as for an homogeneous medium. These curves are inverted to obtain approximate profiles of shear wave velocity and material damping ratio, which are then used to calculate an improved estimate of $Y(r,\omega)$. The subsequent iteration uses the updated $Y(r,\omega)$ to determine more accurate dispersion and attenuation curves. The procedure is repeated until convergence.

Figure 5.5 shows an example of nonlinear regression to determine the complex-valued wavenumber $k^*(\omega)$, hence $V_R(\omega)$ and $\alpha_R(\omega)$. For the sake of a clear graphical representation, the results are reported in terms of phase and amplitude diagrams, but the regression is performed directly on the

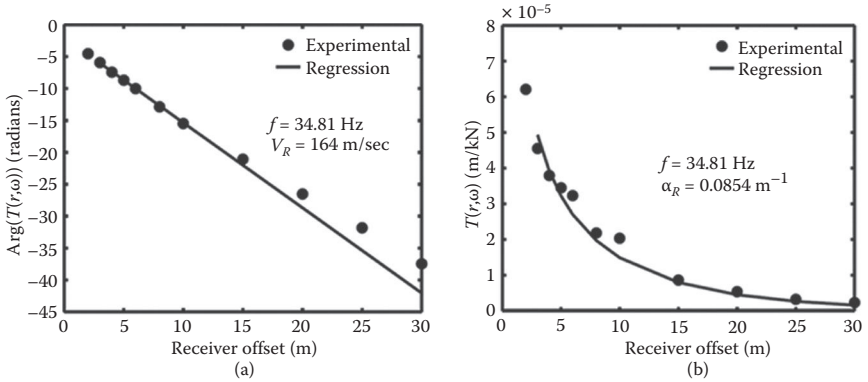

Figure 5.5 Example of regression of displacement transfer function versus offset for coupled estimation of phase velocity and attenuation coefficient: (a) phase versus offset plot; (b) amplitude versus offset plot. (From Rix G. J. et al., *Geotech Test J*, 24, 350–358, 2001.)

complex-valued transfer function. The analysis is repeated at other frequencies to retrieve dispersion and attenuation curves.

Displacement transfer function measurements require an accurate monitoring of the input force. This can be implemented with a load cell for impulsive forces (e.g., sledgehammers) or measuring the acceleration of the armature mass for a controlled source (e.g., a shaker or a Vibroseis) acting in swept-sine mode. Coupling between the sources and the ground represents a significant source of uncertainty. These difficulties can be circumvented by reformulating the transfer function method in terms of deconvolution of the seismic traces.

Deconvolution of a signal $f_2(t)$ with a signal $f_1(t)$ is represented in the frequency domain as the ratio between the Fourier transform of the two signals, $F_2(\omega)$ and $F_1(\omega)$, respectively

$$F_{21}(\omega) = \frac{F_2(\omega)}{F_1(\omega)} = \frac{F_2(\omega)\cdot\overline{F_1(\omega)}}{\left|F_1(\omega)\right|^2} \tag{5.11}$$

The spectrum F_{21} contains information about the interstation phase delay and attenuation and represents the wave propagation between two stations (Dziewonki and Hales 1972). The phase information is entirely in the numerator on the right side of Equation 5.11, which corresponds to the cross-correlation of the two signals, $f_2(t)$ and $f_1(t)$, used in the two-station spectral analysis of surface waves (SASW) test for the determination of the phase velocity (see Section 4.3).

The deconvolved time signal can be evaluated as

$$
\begin{aligned}
f_{21}(t) &= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\left[\frac{F_2(\omega)\cdot\overline{F_1(\omega)}}{\left|F_1(\omega)\right|^2}\right]\cdot e^{i\omega t}\cdot d\omega \\
&= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\left[\frac{A_2(\omega)}{A_1(\omega)}\right]\cdot e^{i\left[\omega t-\phi_2(\omega)+\phi_1(\omega)\right]}\cdot d\omega \qquad (5.12)\\
&= \frac{1}{2\pi}\int_{-\infty}^{+\infty}A(\omega)\cdot e^{-i\phi(\omega)}\cdot e^{i\omega t}\cdot d\omega
\end{aligned}
$$

The function f_{21} represents a signal generated by a δ-impulse source acting at the position of the first receiver and detected at the second receiver (Dziewonki and Hales 1972); therefore, F_{21} is equivalent to the transfer function of the system.

Considering a set of multistation measurements of particle velocity along a straight line on the ground surface, the experimental transfer function $\tilde{F}(r,\omega)$ can be estimated via deconvolution of the whole ensemble of signals

$$\tilde{F}(r,\omega) = F_{1i}(\omega) = \frac{F_i(\omega)}{F_1(\omega)} \qquad (5.13)$$

where $F_i(\omega)$ is the Fourier transform of the ith signal detected at distance r from the source; $F_1(\omega)$ is the Fourier transform of the signal detected by the closest receiver; and $F_{1i}(\omega)$ represents the ith deconvolved signal.

The experimental transfer function can be used in a regression process to estimate the dispersion and attenuation curves of surface waves. Considering Equation 5.8, if the response of the receiver placed at $r = r_1$ is used as the reference trace, the theoretical transfer function can be written as

$$\tilde{F}(r,\omega) = \frac{u_2(r,\omega)}{u_2(r_1,\omega)} = \frac{Y(r,\omega)\cdot e^{-i\Psi(r,\omega)}}{Y(r_1,\omega)\cdot e^{-i\Psi(r_1,\omega)}} \qquad (5.14)$$

Assuming $\Psi(r,\omega)=k^*(\omega)\cdot r$, the implicit dependence of the complex-valued phase angle on the source-to-receiver distance is eliminated and Equation 5.14 becomes

$$\tilde{F}(r,\omega) = \frac{Y(r,\omega)}{Y(r_1,\omega)}\cdot e^{-i\cdot k^*(\omega)\cdot(r-r_1)} \qquad (5.15)$$

where $k^*(\omega)$ is the complex wavenumber (see Equation 5.1).

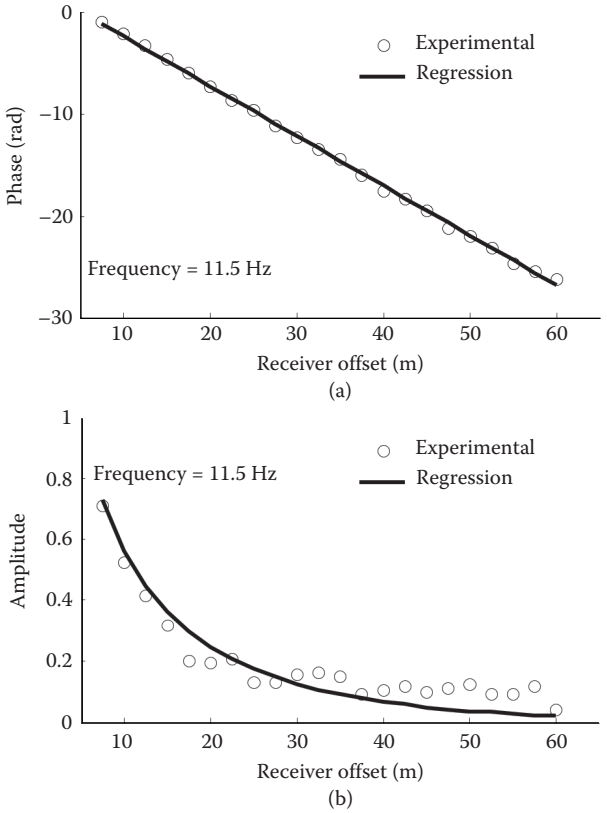

Figure 5.6 Example of regression of deconvolution transfer function versus offset for coupled estimation of phase velocity and attenuation coefficient: (a) phase versus offset plot; (b) amplitude versus offset plot. (From Foti, S., *Geotechnique*, 53, 455–461, 2003.)

The theoretical transfer function of Equation 5.15 is used in a nonlinear regression analysis to estimate the complex wave number $k^*(\omega)$ from the experimental values of the transfer function, obtained by applying Equation 5.13 to experimental data. An example of a comparison between the experimental data and the best fitting theoretical transfer function is reported in Figure 5.6.

5.4 MULTICHANNEL MULTIMODE COMPLEX WAVENUMBER ESTIMATION

The effect of the modal superposition may be severe on the velocity estimation (see Chapter 3) and even more on the attenuation measurement. Indeed, the interference among multiple modes produces amplitude oscillations.

The local gradient of the amplitude, for a wave field resulting from the sum of multiple modes, is not an intrinsic site property, but it also depends on the offset. It is a function of the velocity and of the attenuation of the individual modes and their relative spectral amplitude. As discussed in the previous sections, the Rayleigh geometric spreading function can be introduced to consider the interference, but generally it is not known a priori. Deriving it from the data is not straightforward, and predicting it from the shear wave velocity model introduces uncertainties that are difficult to quantify.

A multichannel method that is able to separate the amplitude contributions of multiple modes and estimate the modal attenuation curves has been recently proposed by Misbah and Strobbia (2014). It estimates the complex wavenumber for multiple linear modes over an array of receivers using multiple sources, if available. The method can be considered an extension of beamforming techniques (see Section 4.6.1), with a complex propagation constant. In particular, the method has been implemented as an extension of the multiple signal classification (MUSIC) method (Schmidt 1986). As for the transfer function method, it can also be applied to nonevenly spaced receiver arrays. The analysis workflow includes the following steps:

1. The cylindrical spreading effect is removed by multiplying each trace by the square root of the source-to-receiver distance. This step removes the geometric spreading contribution, which is not a function of the number of modes and of their interference, from the amplitude decay.
2. Each trace is transformed into the frequency domain.
3. For each frequency
   a. The frequency-dependent autocovariance matrix is calculated as $\mathbf{S}[k] = \overline{\left(\mathbf{T}[k] - \mu_{T[k]}\right)\left(\mathbf{T}[k] - \mu_{T[k]}\right)}$.
   b. The eigenvalues $\lambda[k]$ and eigenvectors $\mathbf{V}[k]$ of the auto covariance matrix $\mathbf{S}[k]$ are evaluated.
   c. The noise eigenvectors $\mathbf{V}_n[k]$ are extracted from the full set of eigenvectors $\mathbf{V}[k]$.
   d. The dot product of the steering vector and each of the noise eigenvectors is calculated to obtain the pseudospectrum. If different shot locations are available, the traces from all shots are analyzed together, estimating the properties of the same set of events, recorded as different signals coming from different sources.
4. The complex wavenumbers for the different modes are evaluated as the values associated with the maxima of the pseudospectrum.

The example presented below refers to a site with a rather simple stratigraphy of fluvial sediments, with interbedded sandy and clayey layers. The raw seismic data are plotted in Figure 5.7.

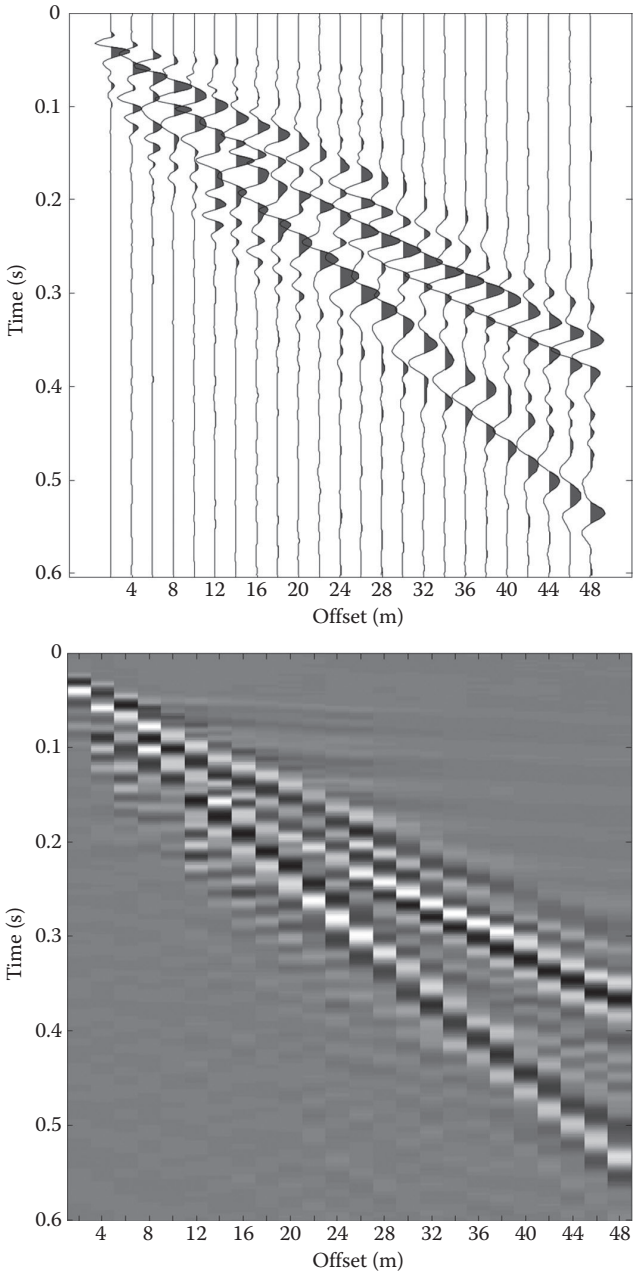

Figure 5.7 Real data seismogram acquired with 24 vertical geophones, spaced 2 m, and with a sledgehammer on plate as source. The data are plotted as wiggle trace (top) and variable density (bottom).

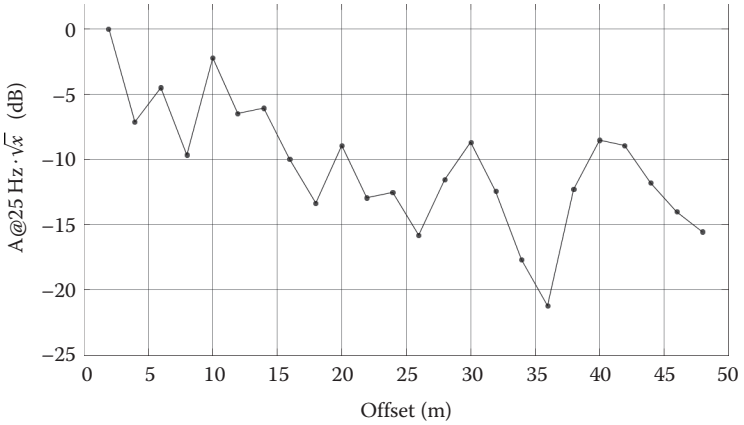

Figure 5.8 Amplitude versus offset, after compensation of the cylindrical geometric spreading, at the frequency of 25 Hz. The amplitude oscillations are due to the modal superposition.

The presence of multiple modes is clear in both representations of the data. A large velocity difference between the slow Rayleigh waves and the fast first arrivals indicates a high Poisson's ratio, which is associated with the presence of a shallow water table. The analysis of the amplitude versus offset shows large oscillations visible in Figure 5.8, where the spectral amplitude at 25 Hz is plotted versus the offset after correction for the cylindrical geometric spreading.

In this case, the amplitude has oscillations due to interference. The number of modes is not so obvious from the inspection of the gather, but it can be assessed looking at the *f–k* spectrum of the gather (Figure 5.9). There are at least three modes in the frequency range 0–50 Hz. The analysis gives the results shown in Figure 5.10.

5.5 OTHER SIMPLIFIED APPROACHES

5.5.1 Half-power bandwidth method

Attenuation curves can be extracted directly from the *f–k* spectra, which are used for the dispersion analysis using an approach derived by the half-power bandwidth (Badsar et al. 2010), usually adopted to evaluate structural damping and damping ratio of soils with the resonant column in the lab. The half-power bandwidth is defined as the difference between frequencies corresponding to an amplitude of the frequency response of a system equal to $1/\sqrt{2}$ the maximum amplitude (Clough and Penzien 1993).

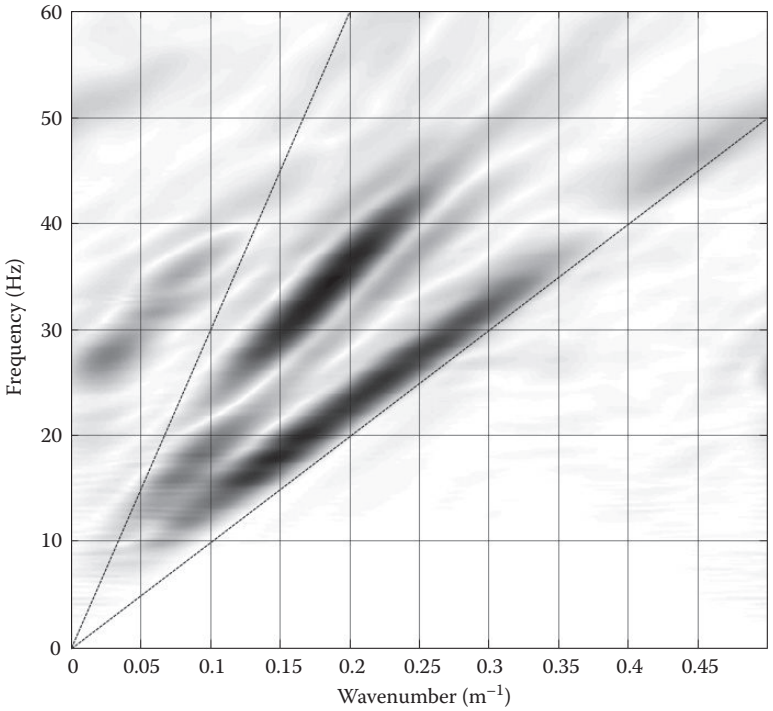

Figure 5.9 Plot of *f–k* spectrum of the gather of Figure 5.7. Multiple modes are present, and at least three modes are visible in the range 20–50 Hz.

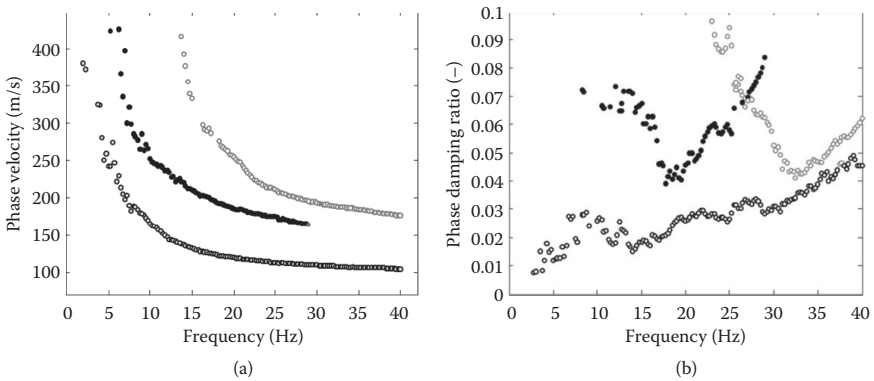

Figure 5.10 (a) Modal phase velocity and (b) modal phase damping ratio for the data of Figure 5.7. Three modes have been identified, and the damping ratio of the fundamental mode ranges between 1% and 4%.

The same concept is applied to the frequency–wavenumber spectrum to obtain the attenuation coefficient of surface waves (Badsar et al. 2010). The main problem in such an approach is due to the influence of spatial sampling; indeed, the limitation of sampling in space associated with the length of the spatial window results in a widening of the spectrum and, hence, in an overestimation of the damping. In order to cope with this issue, Badsar et al. (2010) evaluate the artificial attenuation that is introduced by spatial windowing. True spatial attenuation can be obtained by subtracting the artificial attenuation coefficient from the one estimated on the basis of the half-power bandwidth. The method is able to identify the attenuation coefficient for different modes of propagation, provided that the superposition of their contributions is adequately taken into account.

5.5.2 Spatial decay of the Arias intensity

A method to estimate the damping ratio profile without the intermediate step of estimating of the attenuation curve of Rayleigh waves has been proposed by Badsar et al. (2011). They suggest an inversion for the damping ratio based on the decay of the Arias intensity with offset. Once the shear wave velocity model has been estimated from the inversion of surface wave dispersion, the solution of the forward model is obtained for a linear viscoelastic medium in which the damping ratio is iteratively adjusted until the spatial decay of Arias intensity is close to the experimental value obtained with multistation surface wave testing. The crucial issue for this approach is that the solution of the forward problem requires an accurate elastic model in terms of shear wave velocity and Poisson ratio profiles. Hence, the S- and P-wave velocity profiles need to be evaluated independently and with high accuracy.

5.6 UNCERTAINTY IN THE ATTENUATION MEASUREMENT

The evaluation of the uncertainty of the estimated attenuation follows procedures similar to those described for the velocity (Section 4.8). The procedures based on the fitting allow propagating the data uncertainty onto the estimated parameters.

Some other factors affecting the estimated attenuation are briefly discussed here. The spatial amplitude decay can be related to scattering phenomena: heterogeneities and discontinuities in the subsurface diffract the propagating surface waves, subtracting part of the energy of the direct main wave front which is analyzed. Small-scale features such as inclusions, cavities, cracks, boundaries, and topographic irregularities can reflect a large portion of the energy. For example, the transmission and reflection

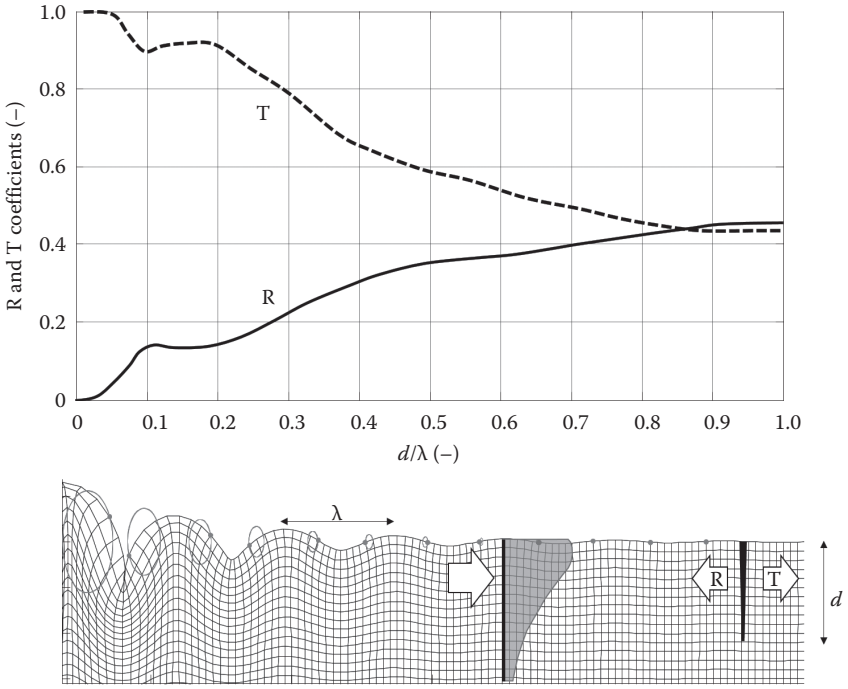

Figure 5.11 Reflection and transmission coefficients for a surface-breaking crack in a homogeneous medium in two-dimensional. (From Rodríguez-Castellanos, A. et al. *Geofis Int* 46(4), 241–248, 2007.)

coefficients for a surface-breaking vertical crack in a homogeneous medium are plotted in two-dimensional (2D) in Figure 5.11. As the ratio of the length of the crack with respect to the wavelength increases, the quantity of reflected energy increases. Such phenomena are not accounted for in the attenuation analysis and may produce a marked overestimation of the intrinsic attenuation and, hence, of the damping ratio of the subsoil.